## Wall time per step: Parse 1M PBMC, 500k (346,259 cells)

Seconds per step. Colour is scaled per step and size: fastest framework 0, slowest 1, on square−root seconds.

| | bixverse mtx | bixverse h5ad | bixverse GPU* | bixverse MLX* | Scanpy mtx | Scanpy h5ad | Scanpy dask | scrapper mtx | scrapper h5ad | scrapper HDF5 | Seurat std mtx | Seurat std h5ad | Seurat BPCells |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| load | 46 | 25 | 21 | 21 | OOM | 12 | 23 | OOM | 19 | 372 | OOM | OOM | 71 |
| qc | 1.8 | 1.8 | 1.2 | 1.2 | – | 3.8 | 4.0 | – | 2.1 | 5.6 | – | – | 4.6 |
| normalise | n/a | n/a | n/a | n/a | – | 3.5 | 0.0 | – | 0.4 | 0.0 | – | – | 2.3 |
| hvg | 1.1 | 1.1 | 0.8 | 0.8 | – | 3.2 | 5.0 | – | 0.9 | 13 | – | – | 14 |
| pca | 2.2 | 2.0 | 1.1 | 0.8 | – | 5.7 | 9.8 | – | 14 | 51 | – | – | 352 |
| harmony | 2.1 | 2.3 | 1.3 | 0.8 | – | 2.4 | 2.3 | – | 87 | 125 | – | – | 89 |
| knn_snn | 12 | 12 | 5.8 | 5.4 | – | 31 | 27 | – | 27 | 29 | – | – | 31 |
| cluster | 4.0 | 4.1 | 3.3 | 3.3 | – | 14 | 13 | – | 85 | 83 | – | – | 104 |
| cluster_fast | 12 | 12 | 1.9 | 1.6 | n/a | n/a | n/a | – | 118 | 119 | n/a | n/a | n/a |
| umap | 22 | 22 | 2.6 | 2.5 | – | 155 | 153 | – | 128 | 127 | – | – | 198 |
| dge | 2.8 | 2.9 | 2.4 | 2.3 | – | 602 | – | – | 39 | 1,401 | – | – | 555 |
| tsne_default | 151 | 154 | 136 | 135 | – | 796 | 806 | n/a | n/a | n/a | – | – | 588 |
| tsne_fft | 39 | 38 | 9.4 | 14 | – | 109 | 109 | n/a | n/a | n/a | – | – | 106 |
| tsne_quick | 24 | 25 | 15 | 14 | n/a | n/a | n/a | – | 84 | 87 | n/a | n/a | n/a |
| total (no tSNE) | 94 | 73 | 39 | 38 | failed | 832 | 237 | failed | 403 | 2,205 | failed | failed | 1,421 |

Black: n/a (step not in that framework), OOM or TIMEOUT where the run died, − not reached.
Each step is scaled over the runs that finished it: at 1M the bixverse CPU tSNE is red because the others timed out.
Scanpy dask has no DGE step. bixverse GPU and MLX run natively without a memory cap.
bixverse CPU tSNE builds its kNN with IVF from 500k (kmknn at 100k). (qd) steps are not in the total.